Dipayan Sarkar

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Research Summary

Currently pursuing a PhD in bioinformatics and computational biology, focused on deep learning methods for biological sequence-based prediction and generative model development. Experience includes neural network training, transfer learning with pretrained protein language models, and analysis of large-scale biological datasets. Current work involves autoregressive generative modeling of biological sequences, with ongoing exploration of diffusion-based approaches for biological data.

Work Experience

University of North Bengal — **Siliguri, India**
PhD Research Scholar — *2022–Present*
Department of Bioinformatics

- Deep learning architectures for sequence-based protein-protein interaction prediction using Attention-based hybrid architecture with transfer learning.
- Sequence based generative modeling of protein binder generation using autoregressive and diffusion-based approaches.
- Development of web-based platforms for model inference and protein interaction network visualization

Education

University of North Bengal — **India**
Ph.D. in Bioinformatics — *Expected March 2027*
Advisor: Dr. Chiranjib Sarkar
Focus: Deep learning-based frameworks for protein-protein interaction network prediction.

University of North Bengal — **India**
M.Sc. in Botany, 77.25% - First Class — *2021*
Specialization: Biochemistry

Ananda Chandra College, University of North Bengal — **India**
B.Sc. (Hons.) in Botany, 63.50% - First Class — *2019*

Publications

2026: **Sarkar, D.**, Bardhan, K., & Sarkar, C. Deep-Interact Studio: An Interactive Deep Learning Model Building Platform for Biomolecular Interaction Prediction. **bioRxiv** preprint. doi:10.64898/2026.07.02.736034

2026: **Sarkar, D.**, & Sarkar, C. MoE-Bind: Guiding De Novo Protein Binder Generation with Sparse Experts. **bioRxiv** preprint. doi:10.64898/2026.06.13.732043

2026: **Sarkar, D.**, & Sarkar, C. ARACoFusion: Uncertainty-aware calibrated deep learning for protein-protein interaction network prediction in *Arabidopsis thaliana*. **bioRxiv** preprint. doi:10.64898/2026.05.22.727120

2025: **Sarkar, D.**, & Sarkar, C. AttnSeq-PPI: Enhancing protein-protein interaction network prediction using transfer learning-driven hybrid attention. ***Biochimica et Biophysica Acta (BBA)—Proteins and Proteomics***, 141102. doi:10.1016/j.bbapap.2025.141102

2022: Sarkar, C., **Sarkar, D.**, Parsad, R., & Mishra, D. Package `EGRNi`. **CRAN**. cran.r-project.org/package=EGRNi

Software & Tools

Deep-Interact Studio: Interactive web platform for building, training, and comparing custom deep learning models for protein–protein, drug–target, RNA–protein, and protein–DNA interaction prediction, with integrated interpretability.
`deepinteract.compbiosysnbu.in/`

AttnSeq-PPI: Deep learning-based protein–protein interaction prediction platform.
`compbiosysnbu.in/attnseqppi/`

Skills

Programming: Python (PyTorch), R, C

Deep Learning: Transformers, attention mechanisms, transfer learning.

Generative AI: Autoregressive, diffusion-, and flow-based generative models for de novo protein design.

Scalable Training: GPU-based training, parallel training on HPC environments

Bioinformatics: Biological sequence analysis, protein-protein interaction prediction, biomolecular network analysis

Data & Viz: NumPy, Pandas, Matplotlib, Seaborn

Deployment: Docker, Flask, FastAPI, Linux server deployment, model-backed web applications

Awards & Certificates

2021: Qualified **CSIR-NET (JRF)** in Life Sciences - June 2021, **All India Rank: 216**

2022: Qualified **GATE** in Life Sciences (XL)

Grants & Computing Resources

2025: **IndiaAI Compute Initiative** - Awarded GPU cloud compute (NVIDIA L4) under the IndiaAI Mission for the project *Deep learning-based framework for protein-protein interaction network prediction* (Project ID: P1-S2025070964, Sep 2025 – Aug 2026). Funded under CSIR-UGC SRF Fellowship.

2025: **Google TRC (TPU Research Cloud) Award** - Granted free access to Google Cloud TPUs (v4, v5e, v6e) for machine learning research.